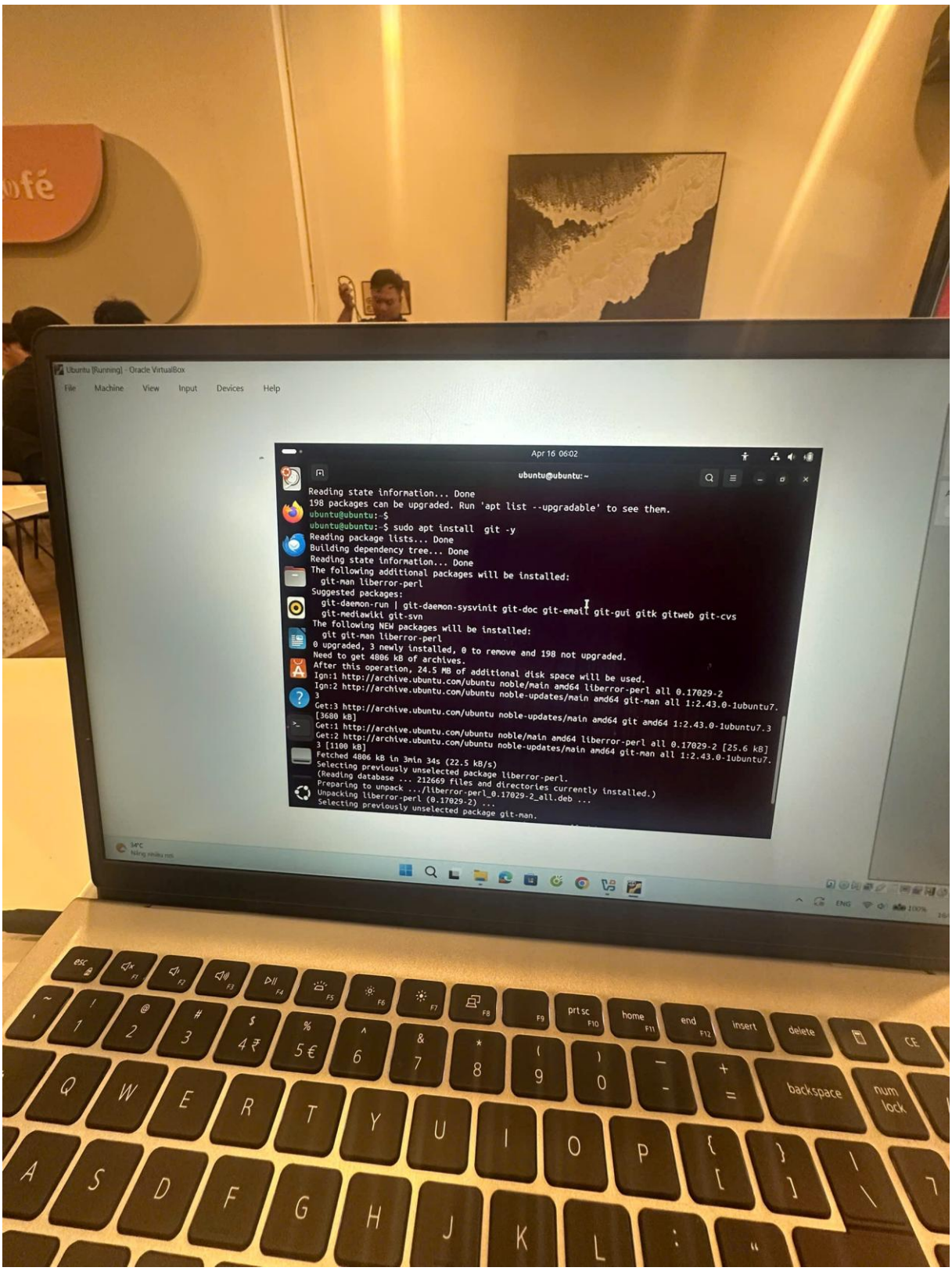
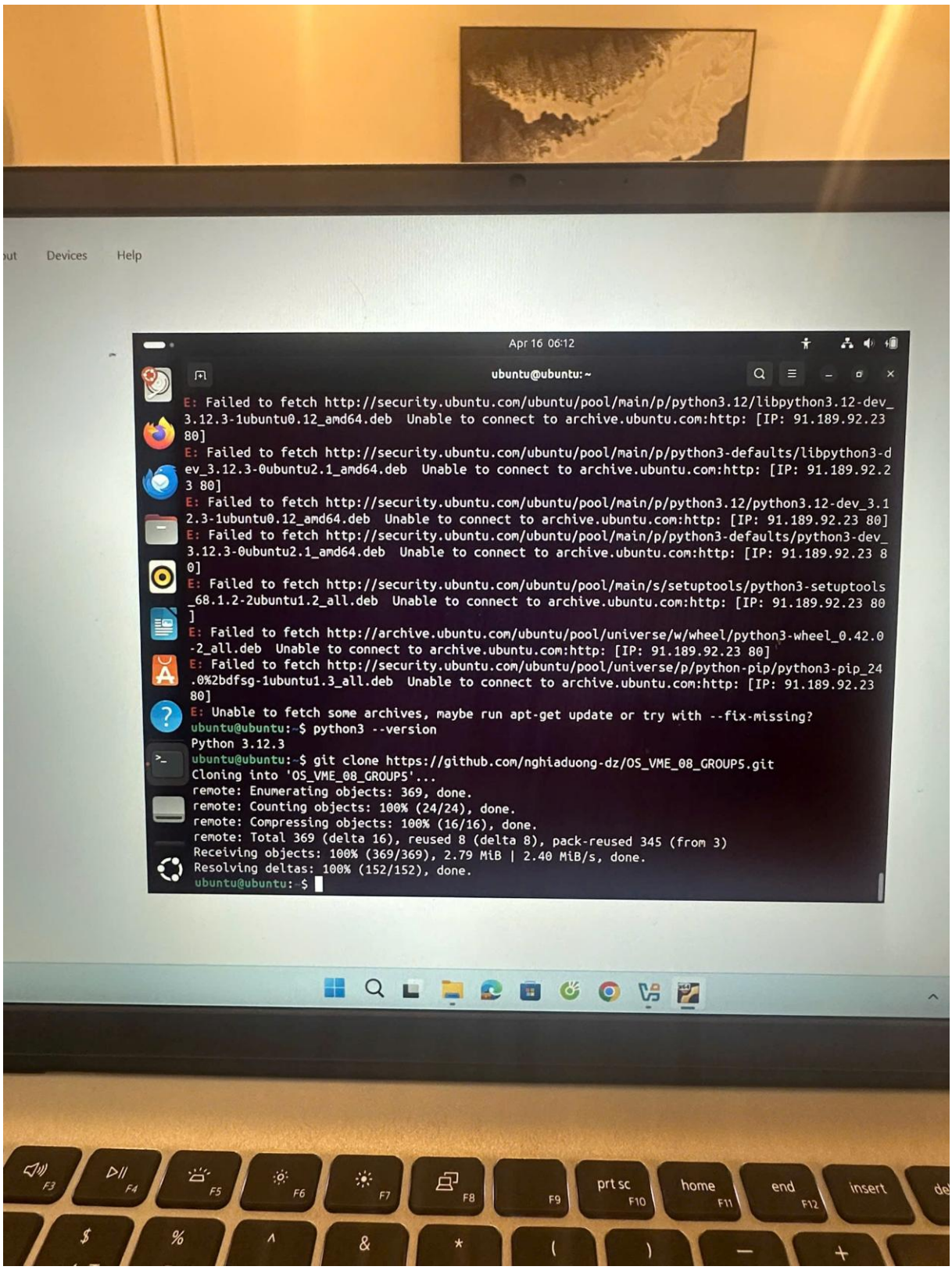
MINH CHỨNG


Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Apr 16 06:02
ubuntu@ubuntu: ~
Reading state information... Done
198 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu:~$
ubuntu@ubuntu:~$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb git-cvs
  git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 198 not upgraded.
Need to get 4806 kB of archives.
After this operation, 24.5 MB of additional disk space will be used.
Ign:1 http://archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl all 0.17029-2
Ign:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1:2.43.0-1ubuntu7.
3
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 git amd64 1:2.43.0-1ubuntu7.3
[3680 kB]
Get:1 http://archive.ubuntu.com/ubuntu noble/main amd64 liberror-perl all 0.17029-2 [25.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 git-man all 1:2.43.0-1ubuntu7.
3 [1100 kB]
Fetched 4806 kB in 3min 34s (22.5 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database ... 212669 files and directories currently installed.)
Preparing to unpack .../liberror-perl_0.17029-2_all.deb ...
Unpacking liberror-perl (0.17029-2) ...
Selecting previously unselected package git-man.


Devices
Help
Apr 16 06:12
ubuntu@ubuntu: ~
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/p/python3.12/libpython3.12-dev_3.12.3-1ubuntu0.12_amd64.deb Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/p/python3-defaults/libpython3-dev_3.12.3-0ubuntu2.1_amd64.deb Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/p/python3.12/python3.12-dev_3.12.3-1ubuntu0.12_amd64.deb Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/p/python3-defaults/python3-dev_3.12.3-0ubuntu2.1_amd64.deb Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/s/setuptools/python3-setuptools_68.1.2-2ubuntu1.2_all.deb Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/universe/w/wheel/python3-wheel_0.42.0-2_all.deb Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/universe/p/python-pip/python3-pip_24.0%2bdfsg-1ubuntu1.3_all.deb Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
ubuntu@ubuntu:~$ python3 --version
Python 3.12.3
ubuntu@ubuntu:~$ git clone https://github.com/nghiaduong-dz/OS_VME_08_GROUP5.git
Cloning into 'OS_VME_08_GROUP5'...
remote: Enumerating objects: 369, done.
remote: Counting objects: 100% (24/24), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 369 (delta 16), reused 8 (delta 8), pack-reused 345 (from 3)
Receiving objects: 100% (369/369), 2.79 MiB | 2.40 MiB/s, done.
Resolving deltas: 100% (152/152), done.
ubuntu@ubuntu:~$

Devices Help

Apr 16 06:14

ubuntu@ubuntu: ~/OS_VME_08_GROUP5/CODE

```
ev_3.12.3-0ubuntu2.1_amd64.deb  Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.2
3 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/p/python3.12/python3.12-dev_3.1
2.3-1ubuntu0.12_amd64.deb  Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/p/python3-defaults/python3-dev_
3.12.3-0ubuntu2.1_amd64.deb  Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 8
0]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/main/s/setuptools/python3-setuptools
_68.1.2-2ubuntu1.2_all.deb  Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80
]
E: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/universe/w/wheel/python3-wheel_0.42.0
-2_all.deb  Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23 80]
E: Failed to fetch http://security.ubuntu.com/ubuntu/pool/universe/p/python-pip/python3-pip_24
.0%2bdfsg-1ubuntu1.3_all.deb  Unable to connect to archive.ubuntu.com:http: [IP: 91.189.92.23
80]
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
ubuntu@ubuntu:~$ python3 --version
Python 3.12.3
ubuntu@ubuntu:~$ git clone https://github.com/nghiaduong-dz/OS_VME_08_GROUP5.git
Cloning into 'OS_VME_08_GROUP5'...
remote: Enumerating objects: 369, done.
remote: Counting objects: 100% (24/24), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 369 (delta 16), reused 8 (delta 8), pack-reused 345 (from 3)
Receiving objects: 100% (369/369), 2.79 MiB | 2.40 MiB/s, done.
Resolving deltas: 100% (152/152), done.
ubuntu@ubuntu:~$ cd OS_VME_08_GROUP5
ubuntu@ubuntu:~/OS_VME_08_GROUP5$ cd CODE
ubuntu@ubuntu:~/OS_VME_08_GROUP5/CODE$ dir
GUI  Output  __pycache__  algorithms  input  main.py  models  readme.md  unit_tests  utils
ubuntu@ubuntu:~/OS_VME_08_GROUP5/CODE$ python main.py
```

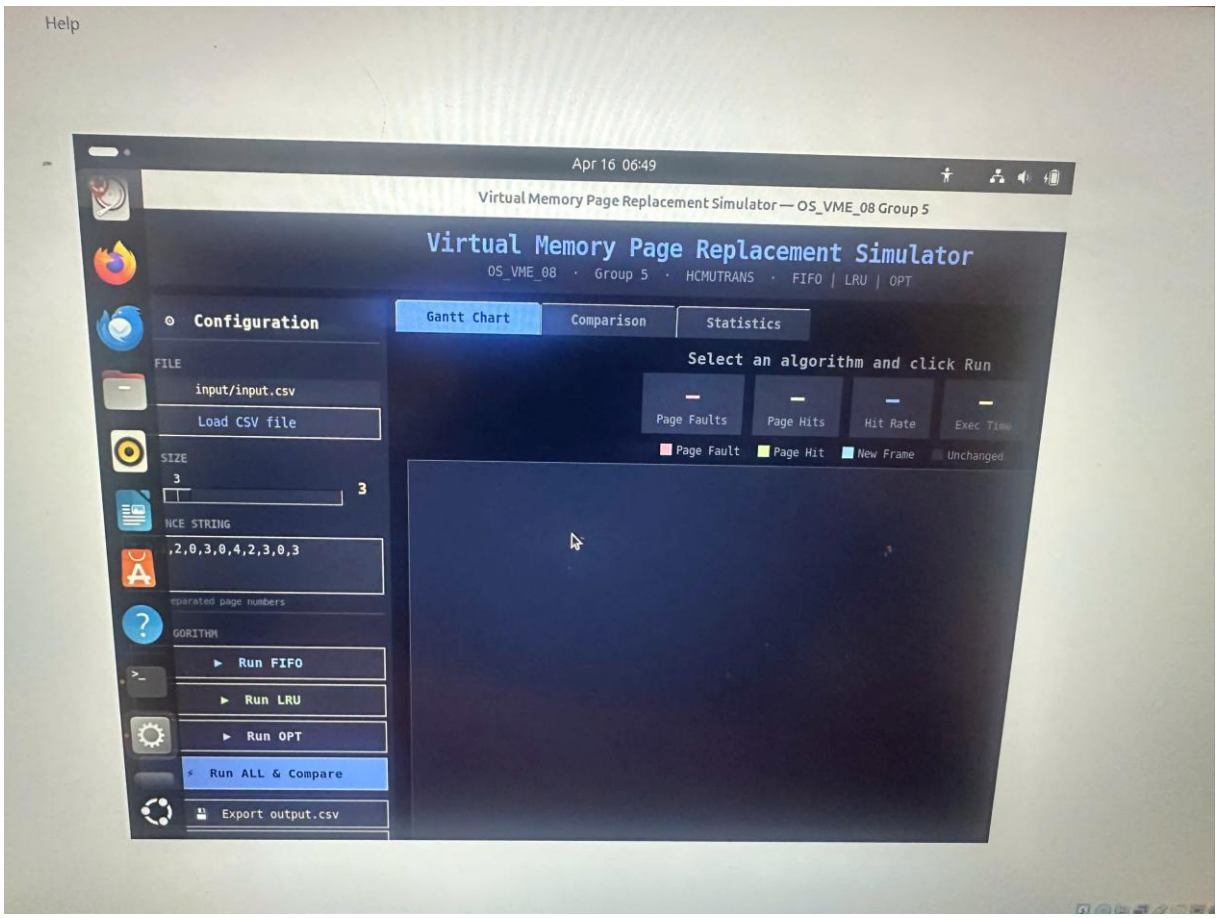
Apr 16 06:26

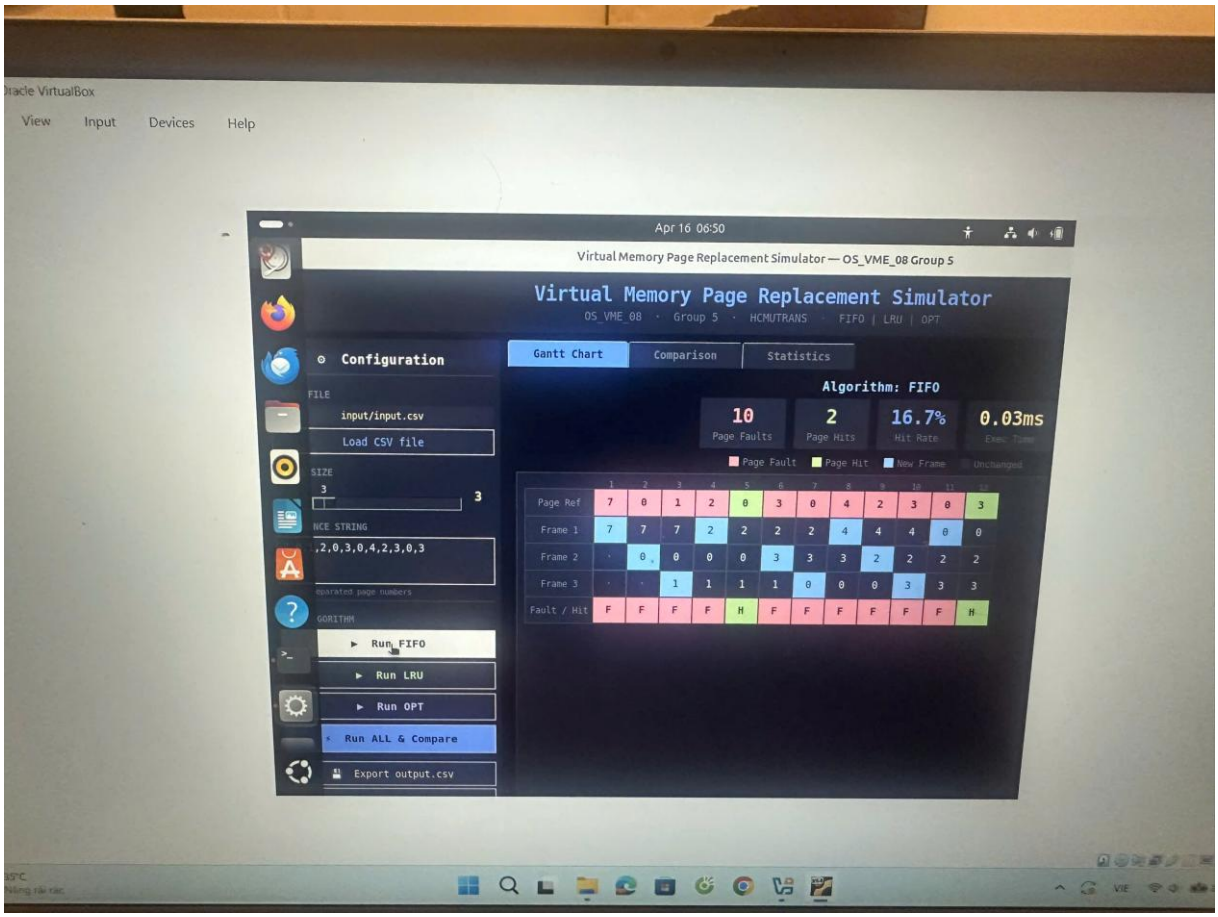
ubuntu@ubuntu: ~

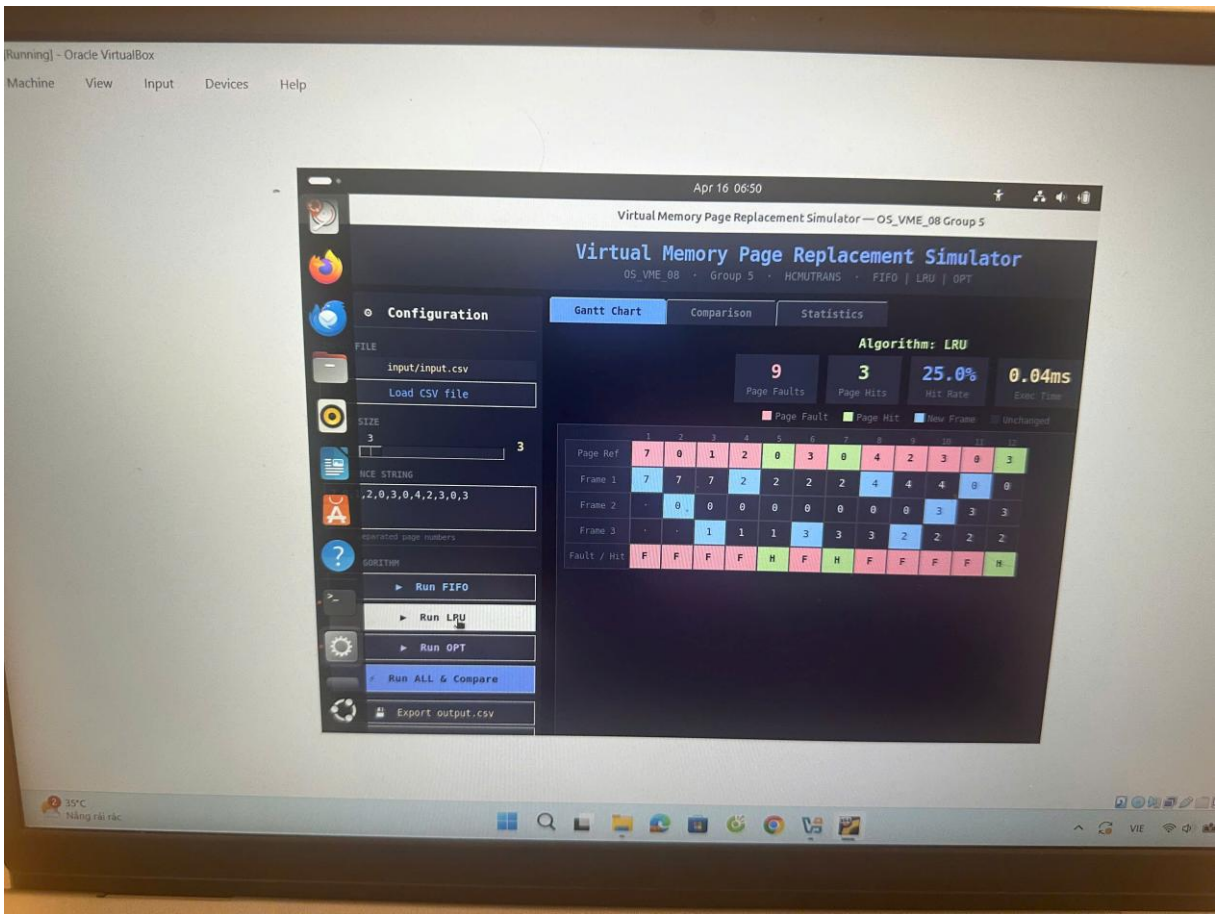
```
Reading state information... Done
198 packages can be upgraded. Run 'apt list --upgradable' to see them.
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/noble-security/InRele
ase  Connection failed [IP: 91.189.91.82 80]
W: Some index files failed to download. They have been ignored, or old ones used
 instead.
ubuntu@ubuntu:~$ git --version
git version 2.43.0
ubuntu@ubuntu:~$ python3 --version
Python 3.12.3
ubuntu@ubuntu:~$ sudo apt install python3-tk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  blt libtk8.6 tk8.6-blt2.5
Suggested packages:
  blt-demo tk8.6 tix python3-tk-dbg
The following NEW packages will be installed:
  blt libtk8.6 python3-tk tk8.6-blt2.5
0 upgraded, 4 newly installed, 0 to remove and 198 not upgraded.
Need to get 1516 kB of archives.
After this operation, 4929 kB of additional disk space will be used.
0% [Waiting for headers]
```

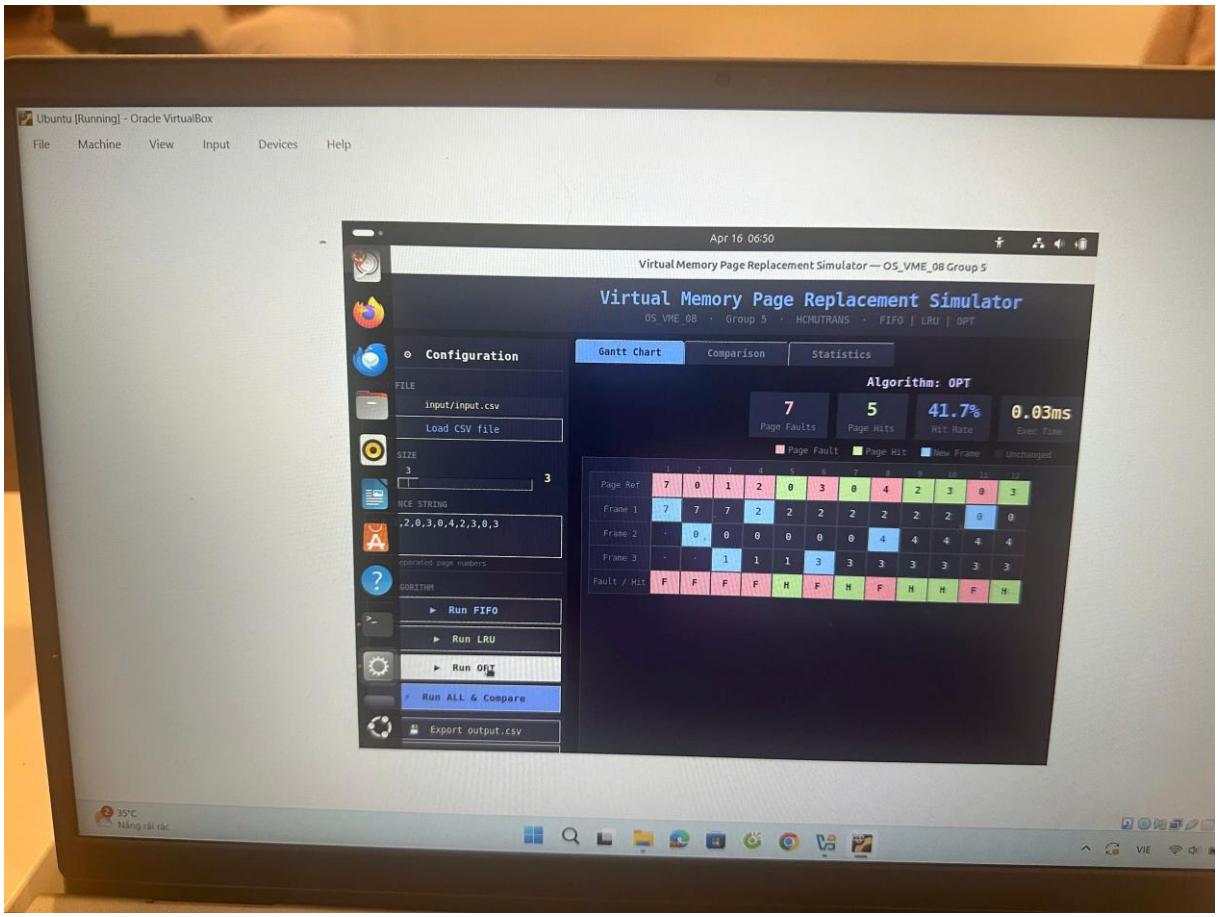
Ubuntu 1.4 LTS

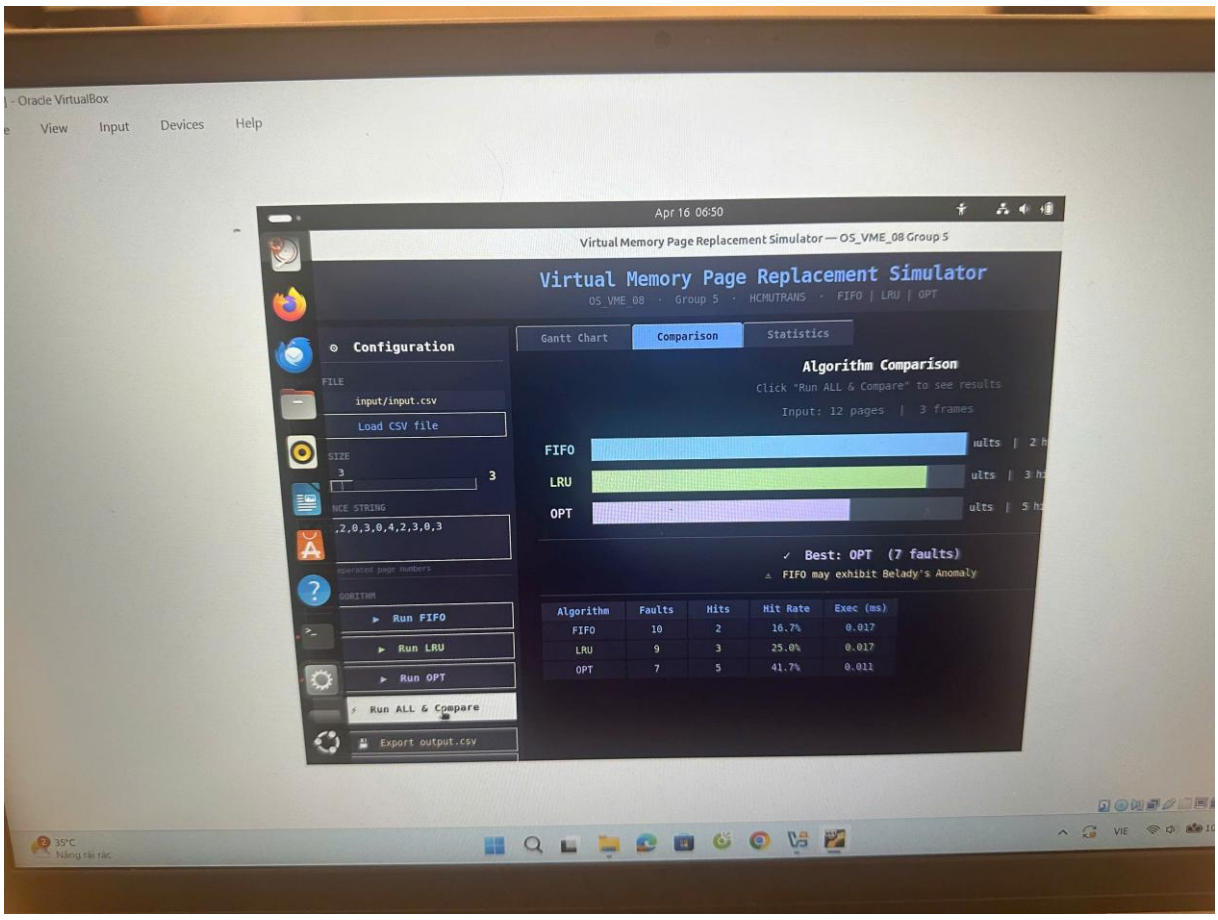
Home


Help
Apr 16 06:49
Virtual Memory Page Replacement Simulator — OS_VME_08 Group 5
Virtual Memory Page Replacement Simulator
OS_VME_08 · Group 5 · HCMUTRANS · FIFO | LRU | OPT
Configuration
Gantt Chart
Comparison
Statistics
FILE
input/input.csv
Load CSV file
SIZE
3
3
NCE STRING
,2,0,3,0,4,2,3,0,3
Select an algorithm and click Run
Page Faults
Page Hits
Hit Rate
Exec Time
Page Fault
Page Hit
New Frame
Unchanged
Run FIFO
Run LRU
Run OPT
Run ALL & Compare
Export output.csv


Oracle VirtualBox
View
Input
Devices
Help
Apr 16 06:50
Virtual Memory Page Replacement Simulator — OS_VME_08 Group 5
Virtual Memory Page Replacement Simulator
OS_VME_08 · Group 5 · HCMUTRANS · FIFO | LRU | OPT
Configuration
Gantt Chart
Comparison
Statistics
Algorithm: FIFO
10
Page Faults
2
Page Hits
16.7%
Hit Rate
0.03ms
Exec Time
Page Fault
Page Hit
New Frame
Unchanged
Page Ref 7 0 1 2 0 3 0 4 2 3 0 3
Frame 1 7 7 7 2 2 2 2 4 4 4 0 0
Frame 2 · 0 0 0 0 3 3 3 2 2 2 2
Frame 3 · · 1 1 1 1 0 0 0 3 3 3
Fault / Hit F F F F H F F F F F F H
FILE
input/input.csv
Load CSV file
SIZE
3
3
NCE STRING
,2,0,3,0,4,2,3,0,3
Run FIFO
Run LRU
Run OPT
Run ALL & Compare
Export output.csv


[Running] - Oracle VirtualBox
Machine View Input Devices Help
Apr 16 06:50
Virtual Memory Page Replacement Simulator — OS_VME_08 Group 5
Virtual Memory Page Replacement Simulator
OS_VME_08 · Group 5 · HCMUTRANS · FIFO | LRU | OPT
Configuration
FILE
input/input.csv
Load CSV file
SIZE
3
3
NCE STRING
,2,0,3,0,4,2,3,0,3
GORITHM
Run FIFO
Run LRU
Run OPT
Run ALL & Compare
Export output.csv
Gantt Chart
Comparison
Statistics
Algorithm: LRU
9
Page Faults
3
Page Hits
25.0%
Hit Rate
0.04ms
Exec Time
Page Fault
Page Hit
New Frame
Unchanged
Page Ref 7 0 1 2 0 3 0 4 2 3 0 3
Frame 1 7 7 7 2 2 2 2 4 4 4 0 0
Frame 2 · 0 0 0 0 0 0 0 0 3 3 3
Frame 3 · · 1 1 1 3 3 3 2 2 2 2
Fault / Hit F F F F H F H F F F F H
35°C
Nắng rải rác
VIE


Ubuntu [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Apr 16 06:50
Virtual Memory Page Replacement Simulator — OS_VME_08 Group 5
Virtual Memory Page Replacement Simulator
OS_VME_08 · Group 5 · HCMUTRANS · FIFO | LRU | OPT
Configuration
FILE
input/input.csv
Load CSV file
SIZE
3
3
NCE STRING
,2,0,3,0,4,2,3,0,3
GORITHM
Run FIFO
Run LRU
Run OPT
Run ALL & Compare
Export output.csv
Gantt Chart
Comparison
Statistics
Algorithm: OPT
7
Page Faults
5
Page Hits
41.7%
Hit Rate
0.03ms
Exec Time
Page Fault
Page Hit
New Frame
Unchanged
Page Ref 7 0 1 2 0 3 0 4 2 3 0 3
Frame 1 7 7 7 2 2 2 2 2 2 2 0 0
Frame 2 · 0 0 0 0 0 0 4 4 4 4 4
Frame 3 · · 1 1 1 3 3 3 3 3 3 3
Fault / Hit F F F F H F H F H H F H
35°C
Nắng rải rác
VIE


- Oracle VirtualBox
View
Input
Devices
Help
Apr 16 06:50
Virtual Memory Page Replacement Simulator — OS_VME_08 Group 5
Virtual Memory Page Replacement Simulator
OS_VME_08 · Group 5 · HCMUTRANS · FIFO | LRU | OPT
Configuration
FILE
input/input.csv
Load CSV file
SIZE
3
3
NCE STRING
,2,0,3,0,4,2,3,0,3
Run FIFO
Run LRU
Run OPT
Run ALL & Compare
Export output.csv
Gantt Chart
Comparison
Statistics
Algorithm Comparison
Click "Run ALL & Compare" to see results
Input: 12 pages | 3 frames
FIFO
LRU
OPT
Best: OPT (7 faults)
FIFO may exhibit Belady's Anomaly
Algorithm Faults Hits Hit Rate Exec (ms)
FIFO 10 2 16.7% 0.017
LRU 9 3 25.0% 0.017
OPT 7 5 41.7% 0.011